

Features

- Step-by-step client configuration is done by the newly installed Streisand server and users can install it using Web browser based instructions.
- Uses SHA-256 checksums for verifying GPG signatures.
- Supports Amazon EC2, DigitalOcean, Google Compute Engine, Linode or Rackspace.

Official website: <https://github.com/StreisandEffect/streisand>

OpenVPN

OpenVPN is a free and open source, cross-platform VPN software application to help systems administrators to create point-to-point and site-to-site connections in routed or bridged configurations and remote access facilities. It adopts a custom security protocol that uses SSL/TLS for key exchange. OpenVPN can traverse firewalls and NAT firewalls, allowing users to bypass gateways and firewalls that might otherwise block connections.

Architecture components

- *Encryption:* OpenVPN uses the OpenSSL library to facilitate data and control connection encryption and authentication. It also uses HMAC packet authentication to provide an additional layer of security to the connection. It supports mbed TLS for better encryption in real-time traffic flow.
- *Authentication:* It offers pre-shared keys, and certificate based and user name/password based authentication.
- *Networking:* It supports both TCP and UDP transport connections, while multiplexing is created on SSL tunnels on a single UCP/TCP port. It can work better with almost all proxy servers and NAT, as well as get through firewalls.
- *Security:* Provides 256-bit encryption through the OpenSSL library, and has the ability to drop root privileges and use mlockall to prevent swapping sensitive data to

the disk. It also supports the use of smart cards through PKCS#11 based cryptographic tokens.

- *Extensibility:* OpenVPN provides advanced features like logging, enhanced authentication, dynamic firewall updates, RADIUS integration, etc.
- The two versions of OpenVPN available are:
- *Community Edition:* Free and open source.
 - *Access Server:* Commercial community edition with add-on features like LDAP integration, SMB server, Web UI management, and add-on tools installation for smart VPN connectivity.

Features

- SSL/TLS for session authentication and IPsec ESP for tunnel transport over UDP.
- Compatible with SSL/TLS, RSA certificates, X.509 PKI, Dynamic Host Configuration Protocol (DHCP), Network Address Translation (NAT) and TUN/TAP virtual devices.
- Fully automated VPN certificate management and provisioning built-in. External PKI is also possible, so you have full control over your existing PKI and use the OpenVPN solution with it.
- Access control rules let you specify which user or group has access to which IP addresses or subnets, and if VPN clients can contact each other or not.
- Built-in authentication system with Web based management. External authentication systems like PAM, LDAP or RADIUS can also be used. The authentication system is extensible but requires programming knowledge (Python).

Official website: <https://openvpn.net/>
Latest version: 2.4.6

PriTunl

PriTunl is an open source VPN server-cum-management panel designed by

BeyondCorp that enables systems administrators to create a cloud VPN with secure encryption, complex site-to-site links and gateway links. It also provides remote access to users in a local network via a Web interface. It makes use of the OpenVPN protocol to run a VPN server, and uses the Let's Encrypt certificate to secure the VPN server and Web GUI. It uses MongoDB at its backend to store all data.

It features up to five authentication layers, a fully customised plug-in integration system, is cross-platform and provides support for OpenVPN clients and AWS VPC networks.

Features

- All source code is available at GitHub to enable developers to re-configure, re-design and develop advanced features, allowing complete transparency and customisation.
- Enables sysadmins to do all configurations using a Web GUI, which is easier than other VPN servers.
- Facilitates end-to-end security between client and server, and uses Google Authenticator for one-step authentication.
- Uses IPsec for site-to-site links and VPC peering.
- Easy integration and configuration with other third party services using the REST API.

Official website: <https://pritunl.com/>
Latest version: 1.29.1827.6

strongSwan

strongSwan is an open source multi-platform IPsec implementation proposed by Andreas Steffen at the University of Applied Sciences in Rapperswil, Switzerland. This VPN solution implements both IKEv1 and IKEv2 key exchange protocols, and uses UDP encapsulation and port floating for NAT-traversal. It supports the Online Certificate Status Protocol, message fragmentation, and modular plugins for crypto-algorithms; it also supports OSCP, CA as well as RSA private keys and